A first Pell root eliminates an index quotient

A complete equation for syntactic ZFC consistency
in fourteen positive variables, sixty-five operations, and degree 150

Research prepared for Vladimir Reshetnikov and ProveIt
AI-assisted construction with independent internal mathematical review

25 October 2026

Abstract

The maintained admissible942 equation in ProveIt has fifteen positive integer unknowns, sixty-six binary arithmetic operations, and exact total degree 157. We remove an index quotient by making the existing first Pell root carry its congruence. The substitution is $\tau = En + k - R$, with one new positive unknown n replacing the two old unknowns h, τ; the factor $k - hE - R$ disappears. A noncircular sign and size argument recovers the omitted positive quotient from every new zero. Canonical parent zeros give new zeros explicitly. The resulting polynomial has fourteen positive unknowns, sixty-five integer-leaf operations, and exact degree 150, with precisely the same four fixed integer coefficients.

A second, independent construction stores old transition targets as reverse-discovery distances. One mixed-radix formula handles old targets, new targets, and the early halt. It shortens the fully specified transition numeral from 436 to 399 base-62 characters without changing any of the 430 actions, 942 indexed tiles, or four coefficient integers.

Composing the new arithmetic proof with the identified, pinned P66/native/refutation calibration gives, in ZFC, equivalence between absence of positive zeros and Con(ZFC). This is an ordinary mathematical proof, not a new end-to-end Lean or Rocq verification. The archive contains the complete one-page specification, all coefficient bytes, literal arithmetic programs, independent finite checks, and domain translations. The result is a strict improvement of the matched parent, not a global minimality claim.

Matched construction	Positive unknowns	Binary operations	Degree
Pinned parent	15	66	157
This construction	**14**	**65**	**150**

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1 Result, interpretation, and comparison

The target is syntactic consistency: no finite formal derivation of a contradiction exists in the effective ZFC presentation used by ProveIt's retained calibration. No consistency assumption is made in proving an equivalence with this statement.

Theorem 1.1 (Main result)**.** *Let $P_{65} \in \mathbb{Z}[J, F, \alpha, z, f, i, T, s, w, n, \eta, \zeta, y, \gamma]$ be the polynomial specified in Sections 2 and 9. Composing the arithmetic reduction proved here with the pinned parent calibration yields*

$$\mathsf{ZFC} \vdash \left[\left(\forall \mathbf{x} \in \mathbb{Z}_{>0}^{14}\right) P_{65}(\mathbf{x}) \neq 0\right] \iff \mathrm{Con}(\mathsf{ZFC}).$$

Its total degree is exactly 150*. Its displayed arithmetic program uses* 65 *binary operations with integer leaves:* 37 *multiplications,* 16 *additions, and* 12 *subtractions.*

The proof has two explicit layers. The new layer is a uniform elementary arithmetic transformation, proved in full below. The inherited layer is the particular parent's native encoding and its K215/syntactic-refutation calibration. Copies of the immediate mathematical contracts and their exact source identities are included. This report does not replace that inherited development with a fresh proof-assistant build.

All repository comparisons in this report use the immutable commit

`02a78a08603cbae248c19c23a75907c6c6bd6443`.

The relevant parent is the complete *admissible942* specialization, not an earlier numerical tuple with the same arithmetic template [1, 2]. The four large integers are preserved pointwise. In particular, no mixture of coefficient ports from different historical native encodings is used.

There are several meaningful size measures. The primary result improves the matched parent's variable count, operation count, and degree simultaneously. It does not minimize any of these among all possible representations. The pinned repository already contains fewer-variable equations with greater degree or cost, as well as direct proof-calculus quartics with small coefficients [4, 7]. Small-variable representation theorems in the literature provide further, different benchmarks [8, 9].

An operation here is one binary $+, -, \times$; a square costs one multiplication. Fixed integer leaves and reuse of earlier registers cost zero operations. This is an arithmetic-circuit measure, not bit complexity: the four fixed integers require 94 689 unsigned bytes in total. Their complete finite recipe is printed rather than hidden behind an unspecified universal-polynomial parameter.

2 The complete polynomial

The independent unknowns, in their evaluator order, are exactly

$$\boxed{J, F, \alpha, z, f, i, T, s, w, n, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.}$$

The four fixed positive integers are β, K, M_C, M_F, defined in Section 9. All other symbols in this section are polynomial abbreviations and are not additional unknowns.

Define

$$q_- = \beta J, \qquad q = q_- + 1, \qquad C = q - F - \alpha, \tag{1}$$
$$X = wq, \qquad Y = sq^3, \qquad E = XY, \tag{2}$$
$$k = \eta + \zeta, \qquad \ell = Ek Y, \qquad c = kY + \eta, \tag{3}$$
$$a = E + Y, \qquad H = 4a + 3, \qquad \Delta = a^2 + H, \tag{4}$$
$$D = X + ac + \gamma H. \tag{5}$$

Then put

$$R = \big[q_-(q - F) + \alpha\big](q^2 - 1) + (M_C + qM_F)J, \tag{6}$$
$$\tau = En + k - R, \tag{7}$$
$$V = f^2(cT - R) - c, \qquad S = i\Delta c^2. \tag{8}$$

The five factors are

$$N_0 = \tau^2 - \ell(\ell + k), \qquad N_1 = D^2 - \Delta c^2, \tag{9}$$
$$N_a = S^2(V^2 - y^2) + y^2, \tag{10}$$
$$N_t = (K + w)C + (q - F) - zq_-, \qquad N_s = \Delta f^2 - S^2. \tag{11}$$

The requested single equation is

$$\boxed{P_{65} := N_0 N_1 N_a N_t N_s - \Delta = 0.} \tag{12}$$

Every exponent involving the fourteen unknowns is a fixed nonnegative integer. Equation (12) is therefore an ordinary polynomial equation over $\mathbb{Z}$. The finite sums, floors, remainders, and powers used later describe fixed coefficients only; none is an additional operation on a Diophantine unknown.

3 The exact inherited interface

The parent P_{66} uses the same definitions except that h, τ are independent positive unknowns and n is absent. It includes

$$N_k = k - hE - R \qquad \text{and} \qquad P_{66} = N_0 N_1 N_a N_k N_t N_s - \Delta.$$

The immediate sources are the current complete equation [1], the full admissible942 proof [2], and the uniform positive P66 argument [3]. The latter is historically presented with another tuple; its uniform arithmetic interface is used only after the current native theorem supplies the complete current specialization.

We use two consequences of that retained calibration:

(i) Every positive zero of the current P_{66} produces a finite syntactic ZFC refutation.

(ii) Every finite syntactic ZFC refutation produces a canonical positive zero with

$$N_0 = N_1 = N_a = N_k = N_t = 1, \qquad f^2 - \Delta i^2 c^4 = 1, \qquad R > 0.$$

These are mathematical source theorems, not assumptions that ZFC is consistent or inconsistent. The new transformation needs only these two directions. It does not need the stronger claim that all parent zeros satisfy the canonical sign conditions.

The new argument uses the following simple scalar properties:

$$\beta \geq 31, \qquad K > 0, \qquad 0 < M_C < \beta < M_F < 2\beta. \tag{13}$$

Here M_F is the *source* mask, including the inherited single β shift. The supplied coefficient checker verifies (13) on the full actual integers and compares their canonical bytes with the pinned certificate.

The parent's native theorem has additional geometric, mask, and controller content. We do not attempt to replace those premises by the scalar boxes. The boxes suffice for the *new arithmetic reduction*; the complete current native theorem supplies the meaning of the fixed coefficients. Since all four integers are unchanged, the parent's finite controller certificate is unchanged too.

4 An elementary Pell lemma

The useful structure is the first norm, before the later norm-combining steps. Its discriminant is the product of consecutive integers.

Lemma 4.1 (Consecutive-product Pell equations)**.** *Let $m \geq 2$ be an integer and let $d = m(m+1)$.*

(a) The equation $x^2 - dy^2 = -1$ has no integer solution.

(b) Every solution of $x^2 - dy^2 = 1$ with $x, y > 0$ has

$$x = \chi_{2m+1}(r), \qquad y = 2\psi_{2m+1}(r), \qquad r \geq 1,$$

where

$$\chi_A(r) + \psi_A(r)\sqrt{A^2-1} = (A + \sqrt{A^2-1})^r.$$

(c) If $E \mid m$, then $x \equiv 1 \pmod{E}$. At $r = 1$ one has $y = 2$; at $r \geq 2$ one has $y \geq 4(2m+1)$. Also $x > y$.

Proof. For (a), a solution has $y \neq 0$. Choose one with least positive $|y|$, and replace x, y by their absolute values. Since $x^2 = dy^2 - 1$, the inequalities

$$my < x < (m + \tfrac{1}{2})y$$

follow by squaring: $my^2 > 1$ and $(m + \frac{1}{2})^2 - d = \frac{1}{4}$. The integer transformation

$$x' = (2m+1)x - 2dy, \qquad y' = (2m+1)y - 2x$$

preserves the norm because $(2m+1)^2 - 4d = 1$. The displayed bounds give $0 < y' < y$. Taking $|x'|$ if necessary produces a smaller solution of norm -1, a contradiction.

For (b), no solution has $y = 1$, since

$$m^2 < d + 1 < (m+1)^2.$$

For $y \geq 2$, norm one implies

$$my < x \leq (m + \tfrac{1}{2})y.$$

The same transformation gives $0 \le y' < y$. Its first coordinate is positive: squaring the two positive terms gives

$$(2m+1)^2x^2 - 4d^2y^2 = dy^2 + 4d + 1 > 0.$$

Descent terminates at $(1,0)$. Reversing it proves that every solution is a positive power of $(2m+1)+2\sqrt{d}$. Since $(2m+1)^2 - 1 = 4d$, this is the claimed parametrization.

For (c), the first coordinate satisfies the integer recurrence

$$\chi_A(0) = 1, \quad \chi_A(1) = A, \quad \chi_A(r+1) = 2A\chi_A(r) - \chi_A(r-1).$$

If $A = 2m+1 \equiv 1 \pmod{E}$, induction gives $\chi_A(r) \equiv 1 \pmod{E}$. The second coordinates begin $\psi_A(1) = 1$, $\psi_A(2) = 2A$ and strictly increase. Finally $x^2 = 1 + dy^2 > y^2$. □

This proof is elementary integer arithmetic and induction. The apparent real square roots merely abbreviate the equivalent pair recurrences. Thus no external-standardness or real-analysis assumption is needed when formalizing it in ZFC.

5 Recovering all unit signs before using the new root

Fix an arbitrary positive zero of P_{65}. We first recover signs without assuming that its polynomial abbreviation τ is positive. This order is essential: τ is no longer an independent positive coordinate.

For every positive assignment,

$$q \ge 32, \quad X, Y, E, a, c > 0, \qquad \Delta = (a+2)^2 - 1 > 0.$$

Moreover,

$$N_s = \Delta\, U_s, \qquad U_s = f^2 - \Delta(ic^2)^2.$$

Cancelling the nonzero Δ in $P_{65} = 0$ gives

$$N_0 N_1 N_a N_t U_s = 1. \tag{14}$$

Each of the five integer factors is consequently $+1$ or -1.

Since $\ell = EkY$, putting $m = EY \ge 2$ gives

$$N_0 = \tau^2 - m(m+1)k^2.$$

Lemma 4.1(a) excludes $N_0 = -1$, even if τ has not yet been shown positive.

The value of Δ is 0 or 3 modulo four. A square minus Δ times a square cannot be 3 modulo four. Therefore

$$N_1 \ne -1, \qquad U_s \ne -1.$$

For the remaining norm, if S is even then $N_a \equiv y^2 \pmod 4$; if S is odd then $N_a \equiv V^2 \pmod 4$. Thus $N_a \ne -1$. Equation (14) now forces the only remaining sign:

$$\boxed{N_0 = N_1 = N_a = N_t = U_s = 1.} \tag{15}$$

The old product had two unresolved linear signs, N_k, N_t. Removing N_k has a useful additional effect: it makes $N_t = 1$ available immediately. This is what permits the following size estimates without assuming the missing quotient in advance.

6 Rank bounds and recovery of the positive quotient

6.1 Bounds derived directly from transport

Suppose $C \le 0$. Since $K + w > 0$, $F, z \ge 1$, and $q_- = q - 1$,

$$N_t = (K + w)C + (q - F) - z(q - 1) \le (q - 1) - z(q - 1) \le 0.$$

This contradicts $N_t = 1$, so $C > 0$.

Write $u = q - F = C + \alpha$. Positivity gives

$$2 \le u \le q - 1, \qquad 1 \le \alpha \le u - 1.$$

Consequently

$$2q - 1 \le G := (q - 1)u + \alpha \le q^2 - q - 1.$$

By the coefficient boxes (13) and $\beta J = q - 1$,

$$0 < (M_C + qM_F)J < (1 + 2q)(q - 1).$$

Substitution into (6) yields

$$q(q^2 - 1) < R < q^4 - q^3. \tag{16}$$

For the upper bound the cancellation is exact:

$$(q^2 - q - 1)(q^2 - 1) + (1 + 2q)(q - 1) = q^4 - q^3.$$

Also $E = wsq^4 \ge q^4$. Thus $R > 1$, $R + 2 < E$, and

$$\tau = En + k - R \ge E + k - R > k + 2 > 0. \tag{17}$$

No part of this argument used an old h coordinate.

6.2 The missing divisibility condition

Apply Lemma 4.1 to $N_0 = 1$. Since $k = \eta + \zeta > 0$ and $\tau > 0$, there is $r \ge 1$ with

$$\tau = \chi_{2EY+1}(r), \qquad k = 2\psi_{2EY+1}(r).$$

As $E \mid EY$, we obtain

$$\tau \equiv 1 \pmod{E}.$$

The definition $\tau = En + k - R$ therefore proves

$$E \mid k - R - 1.$$

Define, in the proof only,

$$h = \frac{k - R - 1}{E}. \tag{18}$$

This is an integer. Since $R + 1 < E$ and $k > 0$, it is greater than -1, hence nonnegative.

It remains to prove strict positivity. If $h = 0$, then $k = R + 1 < E$. At Pell index $r = 1$, however, $k = 2$ would give $R = 1$, contradicting (16). At $r \ge 2$,

$$k \ge 4(2EY + 1) > E,$$

again a contradiction. Therefore $h > 0$, and

$$k - hE - R = 1. \tag{19}$$

Proposition 6.1 (Soundness map)**.** *Every positive zero of P_{65} maps to a positive zero of the pinned P_{66} by retaining the thirteen common coordinates and taking*

$$\tau_{\text{old}} = En + k - R, \qquad h_{\text{old}} = (k - R - 1)/E.$$

Proof. Both inserted coordinates are positive integers by the preceding argument. Every retained abbreviation and norm has exactly its new value, and the restored factor N_k is one. The parent product therefore equals the new product, namely Δ. □

7 Completeness and the internal consistency equivalence

Start with a finite syntactic refutation. The inherited canonical completeness theorem supplies a positive parent zero with $N_k = N_t = 1$, $N_0 = 1$, and $R > 0$. Lemma 4.1 gives

$$\tau_{\text{old}} \equiv 1 \pmod{E}, \qquad \tau_{\text{old}} > k.$$

Since $N_k = 1$, $k - R \equiv 1 \pmod{E}$. Hence

$$n = \frac{\tau_{\text{old}} - k + R}{E} \tag{20}$$

is a positive integer. It satisfies $En + k - R = \tau_{\text{old}}$. All five retained factors have their canonical values, so $P_{65} = 0$.

Combining this construction with Proposition 6.1 and the parent soundness theorem gives the two implications

$$\exists \mathbf{x} \in \mathbb{Z}_{>0}^{14}\ P_{65}(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \exists p\ \text{Refutation}_{\mathsf{ZFC}}(p).$$

Negation proves Theorem 1.1. The proof is a composition of arithmetical statements and explicit witness transformations. It is internal to ZFC: the proof variable p ranges over the theory's natural numbers, rather than an externally selected set of standard codes.

The existence proof does not assert that every parent zero is already canonical. If the retained stronger all-zero sign-recovery theorem is also invoked, (18) and (20) give mutually inverse maps on the full positive zero sets. That stronger conclusion is optional and is not needed for the consistency equivalence.

This argument neither proves nor refutes Con(ZFC). In particular, the certificate checks in the archive do not search for a refutation or for a zero. They authenticate the specified finite arithmetic data.

8 Exact cost and degree

8.1 The arithmetic program

The parent computes N_k with three operations:

$$hE, \qquad k - hE, \qquad k - hE - R.$$

It uses one further multiplication to include N_k in the output. All four operations disappear. The new root costs exactly three:

$$En, \qquad En + k, \qquad En + k - R.$$

The remaining registers can be scheduled in the old dependency order after moving the first norm below the definition of R. Thus

$$66 - 3 - 1 + 3 = \boxed{65}.$$

The literal program in Appendix A has no division, rational constant, dead register, or unused input. Its operator split is 37 multiplications, 16 additions, and 12 subtractions.

8.2 Why the degree is exactly 150

Assign degree one to each unknown and degree zero to each fixed integer. The important degrees are:

Quantity	q	X	Y	E	k	c	ℓ	τ
Degree	1	2	4	6	1	5	11	7

The first norm retains degree 22, since the new τ^2 has degree 14, less than that of ℓ^2.

The main norm requires an actual cancellation calculation. Put $Z = X + \gamma H$. Then

$$N_1 = (ac + Z)^2 - (a^2 + H)c^2 = 2acZ + Z^2 - Hc^2. \tag{21}$$

Its degree is 18, rather than the naive upper bound 22. The five exact factor degrees are

Factor	N_0	N_1	N_a	N_t	N_s
Exact degree	22	18	62	2	46

Their sum is 150, while $\deg \Delta = 12$.

For a direct noncancellation certificate, write $q_0 = \beta J$, $k_0 = \eta + \zeta$. The complete highest homogeneous part is

$$\begin{aligned} L_{150} = {} & 8q_0^{87}\gamma i^4 k_0^{13} w^{12} s^{25} T^2 f^4 \\ & \cdot \big[w(q_0 - F - \alpha) - z q_0\big]. \end{aligned} \tag{22}$$

In particular, the coefficient of

$$J^{88} z \gamma i^4 \eta^{13} w^{12} s^{25} T^2 f^4$$

is $-8\beta^{88} \neq 0$. This proves the exact degree at the actual fixed tuple, rather than only at generic formal coefficient parameters.

The arithmetic checker independently expands the entire main-norm ancestor cone, obtaining 323 terms, and compares all 56 terms of (22). The independent checker also verifies the displayed formula identities and a full univariate specialization of degree 150. The latter is a finite cross-check; the proof of the global upper bound and the actual nonzero coefficient is the argument above.

9 The four fixed integers and their complete finite recipe

9.1 Masks and kernel

Set

$$\rho = 2^{47}, \qquad B = \rho^{4027}, \qquad \beta = B - 1, \qquad G_m(x) = \sum_{j=0}^{m-1} x^j, \qquad \Lambda = \rho^{1992} + \rho^{2998}.$$

Define

$$\Psi(x,t) = x^{969}(x^9 + t^{17})G_8(tx) + t^8 x^{942}(x^{18} + t^{17})G_9(tx) + (2-t)x^{1005}, \tag{23}$$

$$\Phi(x) = \sum_{j=0}^{941} [1 + 2^{10}(1 + w_j + 2^8 n_j + 2^{17} p_j + 2^{25} s_j)]x^j + 2^{10}\Psi(x,2). \tag{24}$$

The indexed tuples (w_j, n_j, p_j, s_j) are specified below. The remaining three constants are

$$M_C = \beta - \rho^{977} - G_{942}(\rho) - \Psi(\rho, 1), \tag{25}$$

$$M_F = B + 3 + (2^{44} - 2)\Lambda + \rho^{960}G_9(\rho) + \rho^{987}G_8(\rho) + \rho^{1005} + 2^{27}\rho^{4022} - \rho^{4004}, \tag{26}$$

$$K = \rho^{18} + \rho^{1005} + \rho^{4036} + \Lambda\Phi(1/\rho). \tag{27}$$

The last expression is an integer. Every exponent in Φ is at most 1005, while both exponents in Λ exceed 1005; distributing the product removes every displayed negative power.

9.2 The exact indexed tile table

Let u_j, v_j, d_j, for $0 \le j < 430$, be the action data decoded in the next subsection. Define, for $d = 0, 1$,

$$h_0(q,b) = 2q + b + 2, \qquad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$

$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. These definitions are finite operations on fixed data. The tile at index i is the tuple $t_i = (w_i, n_i, p_i, s_i)$ in the following table.

Index i	$t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j+1,\ j = 2q + b < 430$	$((1-d_j)r_{0,u_j}, h_0(q,b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b,\ a, b \in \{0,1\}$	$(0, 432a + b, 0, 432a + b)$
$423 + 2r_{d,q} + b$ $q \in \mathcal{S}_d,\ d, b \in \{0,1\}$	$(dr_{d,q}, b, (1-d)r_{d,q}, h_0(q,b))$
$923 + 2a + b,\ a, b \in \{0,1\}$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t,\ t \in \{434, 435, 436\}$	$(1, 438, 1, t)$
$933 + 3a + b,\ a, b \in \{0,1,2\}$	$(a^2, 437, b, 437)$

The supplied finite reconstruction verifies that these formulas assign each index $0, \dots, 941$ exactly once. They retain the actual initializer $t_{928} = (3, 436, 4, 4)$. The symbol h_0 in this finite table is unrelated to the quotient h removed from the polynomial.

9.3 The new closed action decoder

For $b > 0$, use the least nonnegative remainder and write

$$D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b, \qquad A \bmod b = A - b\left\lfloor A/b \right\rfloor.$$

Let $\mathcal{A}$ be the fully printed numeral below. Set

$$N = \mathcal{A} \bmod 2^{636}, \qquad Q = \left\lfloor \mathcal{A}/2^{636} \right\rfloor .$$

For $0 \le j < 430$, and for prefix quantities also at $j = 430$, put

$$\nu_j = D_2(N, j), \qquad \sigma_j = \sum_{k<j} \nu_k, \tag{28}$$

$$r_j = (\sigma_j + 2)^{1-\nu_j}, \qquad P_j = \prod_{k<j} r_k. \tag{29}$$

Thus $\sigma_0 = 0$, $P_0 = 1$; an empty sum is zero and an empty product is one.

For $0 \le q < 215$, $b \in \{0, 1\}$, $j = 2q + b$, define

$$m_q = D_2(N, 430 + q), \qquad \theta_q = \sum_{k<q} m_k, \tag{30}$$

$$F_q = 14 + m_q \left(D_{16}(\lfloor Q/P_{430} \rfloor, \theta_q) - 14 \right), \tag{31}$$

$$u_j = (\sigma_{j+1} - (\lfloor Q/P_j \rfloor \bmod r_j)) \bmod 216, \tag{32}$$

$$v_j = D_2(F_q, 2b), \qquad d_j = D_2(F_q, 2b + 1). \tag{33}$$

Every radix r_j is positive. At a discovery it equals one, and reduction modulo one is well-defined and equals zero. There is no circular dependence: N determines all discovery bits, which determine all prefix products, after which the flag field and targets are decoded.

9.4 The complete 399-character numeral

The digit alphabet, in order of values $0, \ldots, 61$, is

`0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.`

Concatenate the following seven lines into digits $e_0, \ldots, e_{398}$, and set $\mathcal{A} = \sum_{j=0}^{398} e_j 62^{398-j}$.

```
5TPXx0McEgSbYki3sGVNKJLlN8Q3rCXEfmrabEMMQ8k2o7MWvDH3pCkVAr7WqE
SjjWJ6dfq6LXnsZ6OHq7X5iVUPDlwMCAVZSQetoQ0ToXWH6MDnc00LmAYqNDjl
4vOt5Pyqlm1w3QNJbgusSWi8PpjYYL0GfwYX6CiaBwhYAamgcErq0NvxD1RlnG
hk2ZQlIzR5QYJ7qPAGs4LiKjiwLVxBklwY6mc0rXASKtPvKSExJCqmSqOQA4We
dyi3CmNTqaYAAUHXNKVii0Y7dPyuBGK3Wf9xU3vNHYczDPSJUREK7toNXVZLE0
Av8LC0sn6Z6b3e6vj2YEsfHM9lAXImUXzNvBajRskYaVHYhxtRPsiISO0Rst1h
16ymUkkXf1F2lVOfi8rCWgfBlPp
```

This numeral, the displayed decoder, the indexed table, and (23)–(27) completely specify the coefficients. No lookup in the old 436-character numeral is part of the new definition.

9.5 Exact coefficient identities

The coefficients reconstructed by this recipe equal the current admissible942 coefficients as integers, not merely modulo a chosen test modulus.

Coefficient	Bit length	Minimal unsigned bytes
β	189 269	23 659
K	189 693	23 712
M_C	189 269	23 659
M_F	189 270	23 659
Total		94 689

The full byte identities are recorded in `equation/coefficients.json`. The inherited certificate is copied as `sources/admissible942_certificate.json`. Thus the new numeral does not create new controller or native-language obligations.

10 Why reverse-discovery distances shorten the decoder

The repository had already explored mixed-radix target packing [6]. Its smaller numerals did not automatically make a shorter complete description: their decoding overhead could outweigh the saved digits. The new component is the choice of stored digit.

Lemma 10.1 (Reverse-discovery packing)**.** *Let active labels be $0, \dots, H-1$, with halt label H, and suppose m fixed actions discover the active states in order. Write $\nu_j \in \{0,1\}$ for a discovery and $\sigma_j = \sum_{k<j} \nu_k$, with $\sigma_m = H-1$. A discovery targets $\sigma_j + 1$; an old active target is at most σ_j. Define*

$$r_j = (\sigma_j + 2)^{1-\nu_j}, \qquad P_j = \prod_{k<j} r_k.$$

Store the digit

$$d_j = \begin{cases} 0, & \nu_j = 1, \\ \sigma_j - u_j, & u_j \le \sigma_j \text{ is an old active target}, \\ \sigma_j + 1, & u_j = H. \end{cases}$$

Let $T_* = \sum_{j<m} d_j P_j$, and let $Q_* = T_* + P_m E_*$ for any nonnegative auxiliary integer E_*. Then $0 \le T_* < P_m$, $E_* = \lfloor Q_*/P_m \rfloor$, and

$$u_j = (\sigma_{j+1} - (\lfloor Q_*/P_j \rfloor \bmod r_j)) \bmod (H+1).$$

Proof. Each digit satisfies $0 \le d_j < r_j$, including the radix-one discovery case. Induction on the finite mixed-radix expansion gives $0 \le T_* < P_m$. In extracting a digit at position j, all earlier terms sum to less than P_j, while all later terms and the auxiliary field are divisible by $P_j r_j$. Thus the extracted remainder is exactly d_j.

At a discovery the subtraction gives $\sigma_j + 1$; at an old active target it gives u_j; at a halt it gives -1, whose least nonnegative residue modulo $H+1$ is H. These cases are exhaustive. □

For the actual table $m = 430$, $H = 215$. The low field N and the exceptional write/move nibbles are unchanged. Only the target representation changes:

$$\mathcal{A}_{\text{new}} = N + 2^{636}(T_* + P_{430} E_*).$$

The unique halt occurs at action 125, after 80 discoveries. Its stored digit is 81, so the unreduced target is -1. This explains why the outer modulo 216 in (32) is necessary.

The new numeral has 2373 bits and 399 base-62 characters, compared with 2594 bits and 436 characters for its parent. The complete specification, including decoding overhead, also becomes shorter in the delivered template:

Metric	Pinned parent	Combined new page
Numeral characters	436	399
Coefficient recipe, nonwhite extracted characters	1992	1980
Whole page, same convention	2671	2655
Standalone TeX bytes	5025	5014
Pages	1	1
Words in local text extraction	738	743
Locally rebuilt PDF bytes	159 737	160 419

The character convention is `pdftotext -raw`, followed by counting nonwhite characters. It is not a glyph census. The gain includes the inlined auxiliary flag field and the shorter explicit empty-sum/product convention. Word count and PDF byte size increase; no improvement in those metrics is claimed. The repository's stored parent PDF had 159 721 bytes; the table compares both pages rebuilt in the same local environment.

11 Natural and unrestricted integer versions

The witness domain is part of the theorem. Merely allowing arbitrary integers in the fourteen-variable expression would not preserve the argument above.

11.1 Fourteen nonnegative variables

For each old positive variable v, substitute $v = u_v + 1$ with $u_v \in \mathbb{N} = \{0, 1, \ldots\}$. This gives a completely specified polynomial $P_{\mathbb{N}}$ in fourteen nonnegative unknowns. The translation is bijective on domains, so the zero-existence equivalence is immediate. Its degree is exactly 150; adding constants does not alter the highest homogeneous part. A literal evaluator uses $65 + 14 = 79$ operations. It is supplied as `equation/natural14.json`.

11.2 Forty-two unrestricted integer variables

Use, independently for each of the fourteen variables, the substitution

$$v = 1 + a_v^2 + b_v^2 + c_v(c_v + 1), \qquad a_v, b_v, c_v \in \mathbb{Z}. \tag{34}$$

Its value is always positive. It is onto $\mathbb{Z}_{>0}$: for $v \geq 1$, Legendre's three-square theorem represents $4(v-1)+1$ as three integer squares, since that number is not of the exceptional form $4^a(8b+7)$. Reduction modulo four shows exactly one of the squares is odd. Relabeling the coordinates gives

$$4(v-1) + 1 = (2a_v)^2 + (2b_v)^2 + (2c_v + 1)^2,$$

which is equivalent to (34).

Consequently the single ordinary integer equation

$$\boxed{P_{\mathbb{Z}}((a_v, b_v, c_v)_v) := P_{65}\big((1 + a_v^2 + b_v^2 + c_v(c_v+1))_v\big) = 0}$$

has an integer zero exactly when P_{65} has a positive zero. This supplies an unrestricted-integer formulation of the requested consistency equivalence with 42 unknowns.

Each substitution costs seven binary operations, so the literal program uses $65 + 14 \cdot 7 = 163$ operations. Its exact degree is 300. To see that the highest part cannot vanish, take the nonzero L_{150} in (22) and replace each variable by $a_v^2 + b_v^2 + c_v^2$. Setting all b_v, c_v to zero maps distinct old monomials to distinct doubled-exponent monomials in the a_v's; the substitution is injective. The resulting degree-300 form is therefore nonzero.

Witness domain	Unknowns	Operations	Exact degree
Strictly positive integers	14	65	150
Nonnegative integers	14	79	150
Unrestricted integers	42	163	300

The last graph is `equation/integer42.json`. These domain translations are supplied for completeness, not as new low-variable records over $\mathbb{Z}$.

12 Verification, reproducibility, and proof boundaries

12.1 What the finite checks establish

The archive's verifier uses Python 3.10 or later and the standard library. It performs exact integer calculations and sparse polynomial identities. The independent arithmetic checker imports no producer code.

The checks authenticate the pinned parent files; rebuild the three literal equation graphs and all four coefficient files; compare all 430 actions and 942 indexed tiles; verify the complete coefficient integers; check the new scalar interface; count and inspect every arithmetic register; verify the displayed formula identities; and check exact total degree. A separately written decoder uses successive quotient/remainder extraction instead of the author's prefix-floor formula and reconstructs the printed numeral in reverse.

Deliberate changes to the trace sign, a product factor, the witness domain, and three different numeral symbols are rejected. These checks are concrete transcription safeguards. They do not replace the unbounded proofs of Pell classification, the reconstruction of h, or the inherited logical calibration.

The mathematical checks were run normally and with Python optimization enabled. Substantive conditions use explicit failures rather than removable `assert` statements. The normal and optimized results are retained in `verification/`. The one-page equation and this report were compiled, rendered, and visually inspected.

12.2 Reproduction commands

From the extracted archive directory:

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B code/build.py
```

The first two commands are read-only. The third reconstructs the eight equation/coefficient files in memory and compares them byte for byte. To write a fresh reconstruction into a directory that does not exist:

```
python3 -B code/build.py -output-dir /tmp/conzfc-carrier-rebuilt
```

To evaluate a supplied tuple, use a JSON array in the exact coordinate order in the applicable equation graph:

```
python3 -B evaluate.py tuple.json
python3 -B evaluate.py -domain natural tuple.json
python3 -B evaluate.py -domain integer tuple.json
```

Evaluation reports whether the exact value is zero, its sign, bit length, and magnitude hash. The option `-value` prints the entire decimal integer, which can be large. No evaluator performs a root search.

Rebuild the PDFs with a normal TeX installation:

```
pdflatex -interaction=nonstopmode -halt-on-error equation.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The report's last page includes the compiled one-page equation. Its full formula and coefficient recipe also appear in the report source. PDF metadata or font versions can change bytes after rebuilding. Use `verify.py -skip-manifest` after an intentional PDF rebuild; the exact arithmetic and coefficient comparisons remain active.

12.3 What has and has not been newly proved

The new, fully written mathematics consists of the first-root congruence reduction, its noncircular sign and positivity proof, the exact cost and degree calculation, and the reverse-discovery packing lemma. The domain translations use standard number theory. Internal independent AI-assisted reviews are included; they are not external human peer reviews.

The full native-to-refutation correspondence remains the identified ProveIt theorem. The immediate proofs are copied with immutable source identities, but the archive is not a complete Lean/Rocq dependency closure. No new formal-kernel or object-language ZFC derivation has been compiled in this iteration. These distinctions are explicit because a finite coefficient checker cannot by itself establish the unbounded consistency statement.

13 Novelty assessment and the next useful direction

The useful new mechanism is quite specific: preserve the *bare* first Pell root long enough to use the congruence it already carries. The old first norm authenticates $\tau \equiv 1 \pmod{E}$. An affine change in that root turns the congruence into the old quotient condition without a separate h variable or N_k factor. Removing that factor also resolves the transport sign earlier, which makes the positivity proof work.

Source170 already explores related quotient carriers in a combined Pell root and adjusts auxiliary witnesses [5]. The current construction uses the first root itself and does not need that extra scaling. The inspected current formulas and collection summaries did not contain this exact 14/65/150 point. This is a bounded repository novelty assessment, not a literature-wide priority theorem.

The reverse-discovery result likewise builds on an existing repository idea. Mixed-radix compression was already known there; the new digit choice removes a separate correction for the early halt. The exact measured improvement includes the whole decoder, not just its smaller numeral.

Two tempting approaches were examined but not promoted as new results. Dividing the old polynomial by Δ lowers degree but loses a shared multiplication and is already present in earlier reports. Combining two recent direct-quartic pipelines also gave overlapping savings rather than the initially expected new variable record. The delivered result is the independently checked improvement above.

The next promising research question is whether another already authenticated root residue can replace a second quotient *without* introducing a new scaling or sign obligation. The present proof suggests checking such opportunities before combining several Pell roots into one radical norm. That is a research direction, not an additional claimed reduction.

References

[1] ProveIt contributors. *Complete admissible942 equation and current equation metadata*. Pinned repository commit `02a78a08603cbae248c19c23a75907c6c6bd6443`. Complete parent equation; current metadata.

[2] ProveIt contributors. *Removing the forbidden selector: the complete admissible942 tuple*. Pinned full ordinary proof, especially the complete positive witness maps. Copied in `sources/ADMISSIBLE942_PROOF.md`.

[3] ProveIt contributors. *A complete indexed fifteen-positive P66 specialization*. Pinned uniform positive P66 argument. Copied in `sources/P66_POSITIVE_PROOF.md`.

[4] ProveIt contributors. *Con(ZFC) in Diophantine form*. Pinned report collection and comparisons.

[5] ProveIt report170. *Quotient carriers and integral projection for Diophantine consistency*. Pinned report170 README.

[6] ProveIt contributors. *Mixed-radix BFS seed research*. Pinned packing proof and complete-description comparison.

[7] ProveIt report173. *CRT index anchor and direct quartic consistency equations*. Pinned report173.

[8] Y. Matiyasevich and J. Robinson. *Reduction of an arbitrary Diophantine equation to one in 13 unknowns*. Acta Arithmetica 27 (1975). `https://logic.pdmi.ras.ru/~yumat/papers/23_paper/23_home.html`.

[9] Z.-W. Sun. *Ten unknowns for Hilbert's tenth problem over the integers*. arXiv:2609.01594v2, September 2026. `https://arxiv.org/abs/2609.01594`.

A The literal sixty-five-operation evaluator

Each row performs exactly one binary operation. A register is a polynomial abbreviation. The JSON uses `K0` for the fixed integer K; `scaled_Ns` is the displayed N_s. The source records all dependencies explicitly, so repeated subexpressions are reused.

Row	Register	Operation
1	qm	beta * J
2	q	qm + 1
3	q2	q * q
4	q3	q2 * q
5	X	w * q
6	Y	s * q3
7	E	X * Y
8	k	eta + zeta
9	kY	k * Y
10	ell	E * kY
11	c	kY + eta
12	a	E + Y
13	four_a	4 * a
14	H	four_a + 3
15	a2	a * a
16	Delta	a2 + H
17	ac	a * c
18	main_base	X + ac
19	gammaH	gamma * H
20	main_root	main_base + gammaH
21	c2	c * c
22	Dc2	Delta * c2
23	main_square	main_root * main_root
24	N1	main_square - Dc2
25	qF	q - F
26	C	qF - alpha
27	gap_product	qm * qF
28	gap	gap_product + alpha
29	q2minus1	q2 - 1
30	rproduct	gap * q2minus1
31	qMF	q * MF
32	maskfactor	MC + qMF
33	mask	maskfactor * J
34	R	rproduct + mask
35	En	E * n
36	tau_partial	En + k
37	tau	tau_partial - R
38	tau2	tau * tau
39	ell_plus_k	ell + k
40	first_product	ell * ell_plus_k
41	N0	tau2 - first_product
42	Kw	K0 + w
43	transport	Kw * C
44	transport_partial	transport + qF
45	zqm	z * qm
46	Nt	transport_partial - zqm
47	f2	f * f

Row	Register	Operation
48	cT	c * T
49	cTminusR	cT - R
50	f2inner	f2 * cTminusR
51	V	f2inner - c
52	normalized_S	i * Dc2
53	normalized_S2	normalized_S * normalized_S
54	V2	V * V
55	y2	y * y
56	V2minusY2	V2 - y2
57	aux_product	normalized_S2 * V2minusY2
58	Na	aux_product + y2
59	scaled_f2	Delta * f2
60	scaled_Ns	scaled_f2 - normalized_S2
61	product0	N0 * N1
62	product1	product0 * Na
63	product3	product1 * Nt
64	product4	product3 * scaled_Ns
65	polynomial	product4 - Delta

B Complete one-page specification

The following page is the standalone delivered equation. It includes the entire fourteen-variable polynomial, all four constant recipes, the full indexed tile table, the decoder, the alphabet, and all 399 numeral characters. Its small fonts match the pinned comparison template. The preceding sections provide a larger-format account and the proof.

A complete fourteen-variable consistency equation

Strictly positive integer unknowns; exact degree 150; a 65-operation evaluator.

The only unknowns are

$$J, F, \alpha, z, f, i, T, s, w, n, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
q_- = \beta J, \quad q = q_- + 1, \quad C = q - F - \alpha, \\
X = wq, \quad Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \\
\ell = EkY, \quad c = kY + \eta, \quad a = E + Y, \\
H = 4a + 3, \quad \Delta = a^2 + H, \quad D = X + ac + \gamma H, \\
R = [q_-(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
\tau = En + k - R, \quad V = f^2(cT - R) - c, \\
S = i\Delta c^2, \\
N_0 = \tau^2 - \ell(\ell + k), \quad N_1 = D^2 - \Delta c^2, \\
N_a = S^2(V^2 - y^2) + y^2, \\
N_t = (K + w)C + (q - F) - zq_-, \quad N_s = \Delta f^2 - S^2.
\end{gathered}$$

The single equation is

$$\boxed{N_0 N_1 N_a N_t N_s - \Delta = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{47}, \quad B = \rho^{4027}, \quad \beta = B - 1, \\
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1992} + \rho^{2998}.
\end{gathered}$$

$$\begin{aligned}
M_C &= \beta - \rho^{977} \\
&\quad - G_{942}(\rho) - \Psi(\rho, 1), \\
M_F &= B + 3 + (2^{44} - 2)\Lambda \\
&\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1005} \\
&\quad + 2^{27}\rho^{4022} - \rho^{4004}, \\
K &= \rho^{18} + \rho^{1005} + \rho^{4036} \\
&\quad + \Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) &= \sum\nolimits_{i=0}^{941} [1 + 2^{10}(1 + w_i + 2^8 n_i \\
&\quad + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\
\Psi(x, t) &= x^{969}(x^9 + t^{17}) G_8(tx) \\
&\quad + t^8 x^{942}(x^{18} + t^{17}) G_9(tx) \\
&\quad + (2 - t)x^{1005}.
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$\begin{gathered}
h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\}, \\
r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j)r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d)r_{d,q}, h(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
N = \mathcal{A} \bmod 2^{636}, \quad Q = \lfloor \mathcal{A}/2^{636} \rfloor, \\
\nu_j = D_2(N, j), \quad \sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \\
r_j = (\sigma_j + 2)^{1 - \nu_j}, \quad P_j = \prod_{k=0}^{j-1} r_k \ (j \le 430), \\
m_q = D_2(N, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14 + m_q(D_{16}(\lfloor Q/P_{430} \rfloor, \tau_q) - 14), \\
u_j = (\sigma_{j+1} - (\lfloor Q/P_j \rfloor \bmod r_j)) \bmod 216, \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \dots, 61$, is `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`. Concatenate the following 399 characters as digits $e_0, \dots, e_{398}$; put $\mathcal{A} = \sum_{j=0}^{398} e_j 62^{398-j}$.

```
5TPXxOMcEgSbYki3sGVNKJLlN8Q3rCXEfmrabEMMQ8k2o7MWvDH3pCkVAr7WqE
SjjWJ6dfq6LXnsZ6OHq7X5iVUPDlwMCAVZSQetoQOToXWH6MDnc0OLmAYqNDjl
4vOt5Pyqlm1w3QNJbgusSWi8PpjYYLOGfwYX6CiaBwhYAamgcErqONvxD1RlnG
hk2ZQlIzR5QYJ7qPAGs4LiKjiwLVxBklwY6mcOrXASKtPvKSExJCqmSqOQA4We
dyi3CmNTqaYAAUHXNKViiOY7dPyuBGK3Wf9xU3vNHYczDPSJUREK7toNXVZLEO
Av8LCOsn6Z6b3e6vj2YEsfHM9lAXImUXzNvBajRskYaVHYhxtRPsiISOORst1h
16ymUkkXf1F2lVOfi8rCWgfBlPp
```

The ordinary complemented-color, native and Pell arguments give unsolvability iff Con(ZFC). Native calibration and internal-ZFC formalization remain open in Lean. This page is a complete finite specification, not an unsolvability proof.